

BAIRAN CHAKRABORTY

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TECHNICAL SKILLS

Backend: Java · Spring Boot · Spring Security · JWT · Microservices · Spring AI · REST APIs

Frontend & Database: React · JavaScript · HTML/CSS · TensorFlow.js · PostgreSQL · MongoDB

Tools & CS Fundamentals: Git · Postman · Python · SQL · DSA · OOP · DBMS · OS · System Design · Compiler Design

PROJECTS

Custom Language Interpreter & Web Visualizer (BUAS) [GitHub](#)

Java · Lexer / Parser / AST · Compiler Design · JavaScript · HTML/CSS

- Designed and built a custom programming language in Java with a lexer, recursive-descent parser, and AST-based interpreter supporting if-else, while loops, and variable scoping.
- Built a real-time browser visualizer rendering tokens, AST, and execution output side-by-side with descriptive runtime error reporting.

TradeVerse – Stock Trading Platform [GitHub](#)

Java · Spring Boot · Spring Security · JWT · Microservices · Spring AI · JS

- Architected a microservices-based backend with separate auth and trading services; secured via Spring Security JWT and RBAC; APIs cover stock buy/sell, wallet, and transaction history.
- Built a responsive JS frontend and integrated a Spring AI assistant for market query handling.

Plastic Recycling & Logistics Management System [GitHub](#)

Java · Spring Boot · React · TensorFlow.js · Teachable Machine · PostgreSQL · RBAC

- Full-stack system with 4-role RBAC (ADMIN, OPERATOR, SCANNER, INDUSTRY) for plastic classification, batch inventory, and order dispatch via React + Spring Boot.
- Integrated Teachable Machine + TensorFlow.js for real-time client-side plastic classification, blocking low-confidence predictions before batch creation.

E-Commerce Platform (Full Stack) [GitHub](#)

Java · Spring Boot · React · Razorpay · HMAC SHA256 · PostgreSQL · RBAC

- Integrated Razorpay sandbox with HMAC SHA256 backend signature verification; Admin/User RBAC ensures stock updates only after confirmed payment.
- React frontend with role-aware UI and dynamic stock rendering; normalized PostgreSQL schema for products, users, orders, and transactions.

EDUCATION

B.Tech, Computer Engineering — Bharati Vidyapeeth College of Engineering, Pune Expected 2027

· GPA: 9.20 / 10

ACHIEVEMENTS & CERTIFICATIONS

- NPTEL Elite — Design and Analysis of Algorithms | NPTEL — Database Management Systems
- Selected at college level for Smart India Hackathon (SIH) 2025
- Solved 300+ DSA problems across arrays, graphs, dynamic programming, and trees